

Nannuri Pranay Kumar Reddy

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Software engineer building distributed, data-intensive backend systems and agentic AI infrastructure — 3 production microservices at 35K-device scale and the vector-search infra behind Video LiveSearch (Netradyne) · U.S. patent co-inventor

EDUCATION

B.Tech, Electrical Engineering — Indian Institute of Technology (BHU) Varanasi | 2021–2025 | **CPI: 9.19/10**

EXPERIENCE

Data Engineer — Netradyne (Driveri fleet-safety platform), Bengaluru

Jul 2025 – Present

- Designed and shipped the **Jump Detection API**, a real-time telemetry anomaly-detection microservice ingesting streams from **35,000 devices at 10-second intervals** with sub-200 ms latency, designed to scale to 100K+ devices — owned end-to-end, from design and NFRs through FastAPI build, tests, Docker, and production monitoring; later extended to fuel-level jumps.
- Benchmarked **FAISS-HNSW vs exact k-NN in OpenSearch across 10K–1M vectors** (512-dim, 100 iterations, P50/95/99): measured up to **123× ANN speedup** at 1M scale, then used the data to right-size production — migrating the 400K+ doc Video LiveSearch index to exact search, **cutting infrastructure cost at ~0.4 s average latency**.
- Shipped the **waste-management vertical prototype** — Java/Quarkus backend + Angular micro-frontend — **by directing AI coding agents** through a self-designed Claude Code delivery pipeline (research → spec → issue decomposition → execution → pre-merge review).
- Shipped two further microservices end-to-end — a **vehicle status-correction API** and a **fuel-usage anomaly-detection API**: tests, Docker integration, infra provisioning, and QA handoff.
- Owned the internal **video search, review & labelling platform** behind **Video LiveSearch** (industry-first natural-language fleet video search, launched Jan 2026) — shipped the query review & feedback workflows, role-based labelling, KPI computation, and multi-cluster deployment.
- Built and operated **4+ production Airflow (MWAA) data pipelines (DAGs)** — field-KPI monitoring, LiveSearch usage reporting, query-sense recommendations — and debugged cross-region AWS connectivity and data-quality failures in production.
- Kept the API fast as volumes grew by **partitioning its high-write PostgreSQL request/response tables** via Alembic migrations.
- Red-teamed the internal LLM chat-assistant**: documented prompt-injection, tool-introspection, thought-field leakage and SQL-extraction vulnerabilities across multi-turn exploits, feeding hardening work.

Summer Intern — Jindal Steel & Power Ltd. (JSPL)

May – Jun 2024

- Built a cloud-connected IoT energy-monitoring system (ESP32 → ThingSpeak → Node.js/Express) enabling remote industrial monitoring for JSPL's Industry 4.0 initiatives.

OPEN-SOURCE PROJECTS

MCP Net — Hierarchical MCP Discovery & Orchestration github.com/npkriami18/mcp-net · Python · FastAPI · MCP · ADK

- DNS-inspired discovery layer for MCP ecosystems: clients know a single root node and resolve downstream MCP services dynamically over a **Root → Domain → Service resolver graph**, separating routing from execution.
- Standardized routing contract (node type, endpoint, confidence, rationale) keeping multi-hop traversal composable and orchestrator-agnostic.

vessel — Modular AI Agent Harness github.com/npkriami18/vessel · Python

- 20-module AI agent harness built from first principles, implementing the full agent stack: agent loop, tool-use protocol, memory, error recovery, and subagent orchestration.
- Supports git-worktree isolation for safe parallel agent runs and an MCP plugin system for extensible tool integration.

Open-Source Contributor — code-review-graph (19k-star MCP code-intelligence graph for AI coding agents)

github.com/tirth8205/code-review-graph

- Traced silently-reverted commits and restored dropped multi-language parser fixes (Java, GDScript) with refreshed tests — **merged upstream (PR #364)**; two more PRs in review, incl. a Python fix for MCP tool filtering on fastmcp v3+.

PATENT & ACHIEVEMENTS

- Co-inventor — U.S. Patent Application No. 19/567,981** (filed Mar 16, 2026; assigned to Netradyne): natural-language-driven custom alert creation using inertial, vision & video-embedding data on IoT devices — originated from our **2nd-place finish (of 25 teams) at Netradyne Hackathon 2025**.
- FFE–Cognizant Scholar** — merit scholarship with mentorship for exceptional academics.

TECHNICAL SKILLS

Languages: C++, SQL, Python

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, OOP, System Design

Backend: FastAPI, REST API design, microservices, distributed systems

Data Engineering: ETL/data pipelines — Apache Airflow (MWAA), workflow orchestration, PostgreSQL (schema design, partitioning, query optimization, pgvector, PgBouncer), OpenSearch (k-NN, FAISS-HNSW)

Cloud & Infra: AWS (EC2, S3, ASG, ALB, SQS, MWAA), Docker, Consul, observability & alerting (Prometheus, Sentry)

AI & Agents: Agentic AI engineering — LLM evaluation & red-teaming (AI safety), Claude Code (custom skills, prompt engineering, context engineering), MCP (Model Context Protocol), AI orchestration, multi-agent systems, spec-driven development, AI-assisted code review, Google ADK